UltraXAI — Project Update Report

Explainable AI for Breast Cancer Detection from Ultrasound
Developed by: Abdul Rafay — Final Year Project (FYP)
Date: 24 October 2026

1. Summary

This update adds data augmentation to expand the training dataset, fixes a data-leakage bug that inflated earlier accuracy, adds modern training techniques to improve real accuracy, makes the code compatible with Albumentations 2.x, and provides a one-command training pipeline for a local GPU.

2. Data Augmentation & Dataset Volume

New script `training/data/augment_offline.py` writes N augmented copies of every training image and its mask. With 5 copies the training set grows 6×. Only the training split is augmented, so validation/test images never leak into training.

Augmentation	Purpose
Flips, shift / scale / rotate (Affine)	Probe position and orientation changes
Elastic & grid distortion	Tissue deformation, probe pressure
Speckle (multiplicative) & Gaussian noise, blur	Ultrasound noise variety
Brightness / contrast, gamma, CLAHE	Different machine gain settings
Sharpen, coarse dropout	Edge robustness, learn from multiple regions

3. Accuracy Improvements (train_classifiers.py)

Technique	Benefit
Stratified train / val split	Fixes bug: validation used the same images as training, so old scores were too high
Offline-augmented data (--aug_dirs)	More, more varied training data
Label smoothing 0.1	Better calibrated, less over-confident
Mixup (alpha 0.2)	Strong regularisation for small datasets
Warmup + cosine learning rate	Stable fine-tuning of pretrained models
AdamW weight decay 1e-4, gradient clipping	Less overfitting, stable training
Mixed precision (AMP)	Faster GPU training
Flip test-time augmentation	More robust predictions
Save-best fix	Model is saved even if first F1 is 0

4. Compatibility Fixes

Updated to the Albumentations 2.x API: ShiftScaleRotate → Affine, GaussNoise(var_limit) → std_range, new CoarseDropout ranges, removed OpticalDistortion shift_limit, and torch.amp.GradScaler.

5. One-Command Pipeline

`run_full_training.py`: (1) stratified 70/15/15 split → (2) augment train only → (3) train ResNet50 + EfficientNetV2 → (4) save weights to backend/weights/ for the app.

```
python download_busi.py
python run_full_training.py --data_dir datasets/BUSI --copies 5 --epochs 80
```

6. Verification

The full pipeline was tested end-to-end (split, augment, 1 epoch per model, weights saved) on synthetic images with no errors or warnings. Final accuracy on the real BUSI dataset will be reported after GPU training. Note: scores are expected to be lower than before but honest, because the leakage bug is fixed.

7. Files Changed

File	Change
training/data/augment_offline.py	New — dataset expansion
training/data/transforms.py	New augmentations, Albumentations 2.x
training/train_classifiers.py	Split fix + accuracy techniques
run_full_training.py	New — one-command pipeline
README.md	Augmentation, accuracy, changelog, author credit

UltraXAI — Developed by Abdul Rafay